Describe definition and common commands

Kubernetes Commands

1. **kubectl get**: Lists resources such as pods, services, deployments, etc.
    - Usage: kubectl get pods
2. **kubectl describe**: Shows detailed information about a specific resource.
    - Usage: kubectl describe pod [pod-name]
3. **kubectl apply**: Applies a configuration to a resource by filename or stdin.
    - Usage: kubectl apply -f [filename]
4. **kubectl delete**: Deletes resources by filenames, stdin, resources, and names, or by resources and label selector.
    - Usage: kubectl delete pod [pod-name]

Terraform Commands

1. **terraform init**: Initializes a Terraform working directory.
    - Usage: terraform init
2. **terraform plan**: Creates an execution plan, showing what actions Terraform will take.
    - Usage: terraform plan
3. **terraform apply**: Applies the changes required to reach the desired state of the configuration.
    - Usage: terraform apply
4. **terraform destroy**: Destroys the Terraform-managed infrastructure.
    - Usage: terraform destroy

Google Cloud Build (GCB) Commands

1. **gcloud builds submit**: Submits a build using the Google Cloud Build service.
    - Usage: gcloud builds submit --tag gcr.io/[PROJECT-ID]/[IMAGE-NAME]
2. **gcloud builds list**: Lists all builds in the specified project.
    - Usage: gcloud builds list
3. **gcloud builds log**: Shows logs for a specific build.
    - Usage: gcloud builds log [BUILD-ID]

Jenkins Commands

1. **jenkins-cli.jar**: A command-line interface to interact with Jenkins.
    - Usage: java -jar jenkins-cli.jar -s [JENKINS_URL] [command]
2. **build**: Triggers a new build.
    - Usage: java -jar jenkins-cli.jar -s [JENKINS_URL] build [JOB_NAME]
3. **console**: Outputs the console output of a specific build.
    - Usage: java -jar jenkins-cli.jar -s [JENKINS_URL] console [JOB_NAME] [BUILD_NUMBER]

Docker Commands

1. **docker build**: Builds an image from a Dockerfile.
    - Usage: docker build -t [image-name] .
2. **docker run**: Runs a command in a new container.
    - Usage: docker run [image-name]
3. **docker ps**: Lists all running containers.
    - Usage: docker ps

4. **docker stop**: Stops one or more running containers.
    - Usage: docker stop [container-id]

Google Cloud Platform (GCP) Commands

1. **gcloud init**: Initializes gcloud and sets up configurations.
    - Usage: gcloud init
2. **gcloud compute instances create**: Creates a new Compute Engine instance.
    - Usage: gcloud compute instances create [instance-name] --zone=[zone]
3. **gcloud projects list**: Lists all the projects associated with the Google Cloud account.
    - Usage: gcloud projects list
4. **gcloud config set**: Sets a configuration value.
    - Usage: gcloud config set project [project-id]

These commands provide a starting point for interacting with these tools, enabling effective management of infrastructure, application deployment, and cloud resources.

Certainly! Below are the definitions and common commands for a few popular tools and utilities that you might find useful:

1. Command Line Interface (CLI)

Definition: A text-based interface used to interact with software and operating systems by typing commands.

Common Commands:

- `cd [directory]`: Change the current directory.

- `ls` (Unix) / `dir` (Windows): List files and directories in the current directory.

- `cp [source] [destination]`: Copy files or directories.

- `mv [source] [destination]`: Move or rename files or directories.

- `rm [file]`: Remove files or directories.

- `mkdir [directory]`: Create a new directory.

- `pwd`: Print the current working directory.

2. Git

Definition: A version control system for tracking changes in source code during software development.

Common Commands:

- `git init`: Initialize a new Git repository.

- `git clone [repository]`: Clone a repository into a new directory.

- `git status`: Show the working tree status.

- `git add [file]`: Add file contents to the index.

- `git commit -m "[message]"`: Record changes to the repository.

- `git push`: Update remote references along with associated objects.

- `git pull`: Fetch from and integrate with another repository or a local branch.

3. Docker

Definition: A platform for developing, shipping, and running applications inside containers.

Common Commands:

- `docker build [options] [path]`: Build an image from a Dockerfile.

- `docker run [options] [image]`: Run a command in a new container.

- `docker ps`: List running containers.

- `docker stop [container]`: Stop a running container.

- `docker rm [container]`: Remove a container.

- `docker images`: List images.

- `docker pull [repository]`: Pull an image from a repository.

4. Kubernetes

Definition: An open-source system for automating the deployment, scaling, and management of containerized applications.

Common Commands:

- `kubectl get [resources]`: List resources.

- `kubectl describe [resource] [name]`: Show detailed information about a resource.

- `kubectl apply -f [file]`: Apply a configuration to a resource by filename or stdin.

- `kubectl delete [resource] [name]`: Delete resources by filenames, stdin, resources, and names.

- `kubectl logs [pod]`: Print the logs for a container in a pod.

- `kubectl exec [pod] -- [command]`: Execute a command in a container.

5. SQL (Structured Query Language)

Definition: A domain-specific language used in programming and designed for managing data held in a relational database management system.

Common Commands:

- `SELECT [columns] FROM [table]`: Query data from a database.

- `INSERT INTO [table] ([columns]) VALUES ([values])`: Insert data into a table.

- `UPDATE [table] SET [column=value] WHERE [condition]`: Update data in a table.

- `DELETE FROM [table] WHERE [condition]`: Delete data from a table.

- `CREATE TABLE [table] ([columns])`: Create a new table.

- `DROP TABLE [table]`: Delete a table.

6. Python

Definition: A high-level, interpreted programming language known for its readability and versatility.

Common Commands:

- `print([value])`: Print the specified value to the console.

- `def [function_name]([parameters]):`: Define a new function.

- `import [module]`: Import a module.

- `for [variable] in [iterable]:`: Start a for loop.

- `if [condition]:`: Start an if statement.

- `while [condition]:`: Start a while loop.

- `class [ClassName]:`: Define a new class.

Common Commands and Tools

GCP Core Services:

- **Compute**: `gcloud compute instances create`, `gcloud compute instances list`

- **Storage**: `gsutil cp`, `gsutil ls`

- **Networking**: `gcloud compute networks create`, `gcloud compute firewall-rules create`

- **Databases**: `gcloud sql instances create`, `gcloud sql databases create`

- **Containers (GKE)**: `gcloud container clusters create`, `kubectl apply -f`

- **Security**: `gcloud iam roles create`, `gcloud kms keys create`

- **Identity**: `gcloud iam service-accounts create`, `gcloud iam roles list`

Infrastructure as Code (Terraform):

- **Initialization**: `terraform init`

- **Plan**: `terraform plan`

- **Apply**: `terraform apply`

- **Destroy**: `terraform destroy`

Deployment Tools:

- **GitHub**:

  - Clone Repository: `git clone <repo-url>`

  - Branching: `git branch <branch-name>`, `git checkout <branch-name>`, `git merge <branch-name>`

  - Push Changes: `git add .`, `git commit -m "commit message"`, `git push origin <branch-name>`

- **Ansible**:

  - Run Playbook: `ansible-playbook <playbook.yml>`

  - List Hosts: `ansible all --list-hosts`

- **Jenkins**:

  - Trigger Build: `curl -X POST http://<jenkins-url>/job/<job-name>/build`

Programming Languages:

- **Java**:

  - Compile: `javac <filename.java>`

  - Run: `java <classname>`

- **Python**:

  - Run Script: `python <script.py>`

  - Install Packages: `pip install <package-name>`

Linux:

- **Basic Commands**: `ls`, `cd`, `pwd`, `cp`, `mv`, `rm`

- **File Permissions**: `chmod`, `chown`

- **System Monitoring**: `top`, `htop`, `df`, `du`

- **Networking**: `ifconfig`, `netstat`, `ping`, `traceroute`

Definition

GCP certifications validate expertise in various aspects of Google Cloud Platform and related tools. These certifications demonstrate proficiency in designing, developing, and managing robust, secure, scalable, and highly available applications using GCP technologies. Additionally, they cover best practices for cloud security, DevOps, and infrastructure as code (IaC) using Terraform.

Common GCP Certifications:

1. **Google Cloud Certified - Associate Cloud Engineer**: Demonstrates the ability to deploy applications, monitor operations, and manage enterprise solutions.

2. **Google Cloud Certified - Professional Cloud Architect**: Validates skills in designing, developing, and managing solutions on GCP.

3. **Google Cloud Certified - Professional Cloud Developer**: Focuses on building scalable and highly available applications using Google-recommended practices and tools.

4. **Google Cloud Certified - Professional Cloud Security Engineer**: Covers the design and implementation of secure infrastructures on GCP.

5. **Google Cloud Certified - Professional Cloud DevOps Engineer**: Emphasizes efficient development operations that balance service reliability and delivery speed.

Terraform Certification:

- **HashiCorp Certified: Terraform Associate**: Validates understanding of the basic concepts and skills associated with open-source HashiCorp Terraform.

Common Commands and Tools

GCP Certifications Related Commands:

1. **Associate Cloud Engineer**:

   - **Compute**:

     - Create instance: `gcloud compute instances create [INSTANCE_NAME]`

     - List instances: `gcloud compute instances list`

   - **Storage**:

- Create bucket: `gsutil mb gs://[BUCKET_NAME]/`
- List buckets: `gsutil ls`

- **Networking**:
  - Create network: `gcloud compute networks create [NETWORK_NAME]`
  - List firewall rules: `gcloud compute firewall-rules list`

2. **Professional Cloud Architect**:
   - **IAM**:
     - Create service account: `gcloud iam service-accounts create [SERVICE_ACCOUNT_NAME]`
     - List roles: `gcloud iam roles list`
   - **Resource Management**:
     - Create project: `gcloud projects create [PROJECT_ID]`
     - Set project: `gcloud config set project [PROJECT_ID]`

3. **Professional Cloud Developer**:
   - **App Engine**:
     - Deploy app: `gcloud app deploy`
     - List services: `gcloud app services list`
   - **Cloud Functions**:
     - Deploy function: `gcloud functions deploy [FUNCTION_NAME] --runtime [RUNTIME] --trigger-http`
     - List functions: `gcloud functions list`

4. **Professional Cloud Security Engineer**:
   - **Cloud KMS**:
     - Create keyring: `gcloud kms keyrings create [KEYRING_NAME] --location [LOCATION]`

- Create key: `gcloud kms keys create [KEY_NAME] --location [LOCATION] --keyring [KEYRING_NAME] --purpose encryption`

- **Security Command Center**:

  - Enable service: `gcloud services enable securitycenter.googleapis.com`

  - List findings: `gcloud scc findings list`

5. **Professional Cloud DevOps Engineer**:

- **Cloud Build**:

  - Submit build: `gcloud builds submit --tag gcr.io/[PROJECT_ID]/[IMAGE_NAME]`

  - List builds: `gcloud builds list`

- **Cloud Monitoring**:

  - List metrics: `gcloud monitoring metrics list`

  - Create uptime check: `gcloud monitoring uptime-checks create http [CHECK_NAME] --host [HOST] --path [PATH]`

Terraform Associate Certification Commands:

- **Initialization**: `terraform init`

- **Plan**: `terraform plan`

- **Apply**: `terraform apply`

- **Destroy**: `terraform destroy`

- **Format Code**: `terraform fmt`

- **Validate Configuration**: `terraform validate`

- **Show State**: `terraform show`

- **List Resources**: `terraform state list`

These certifications and tools ensure proficiency in deploying, managing, and securing cloud infrastructure and applications using GCP and Terraform.